

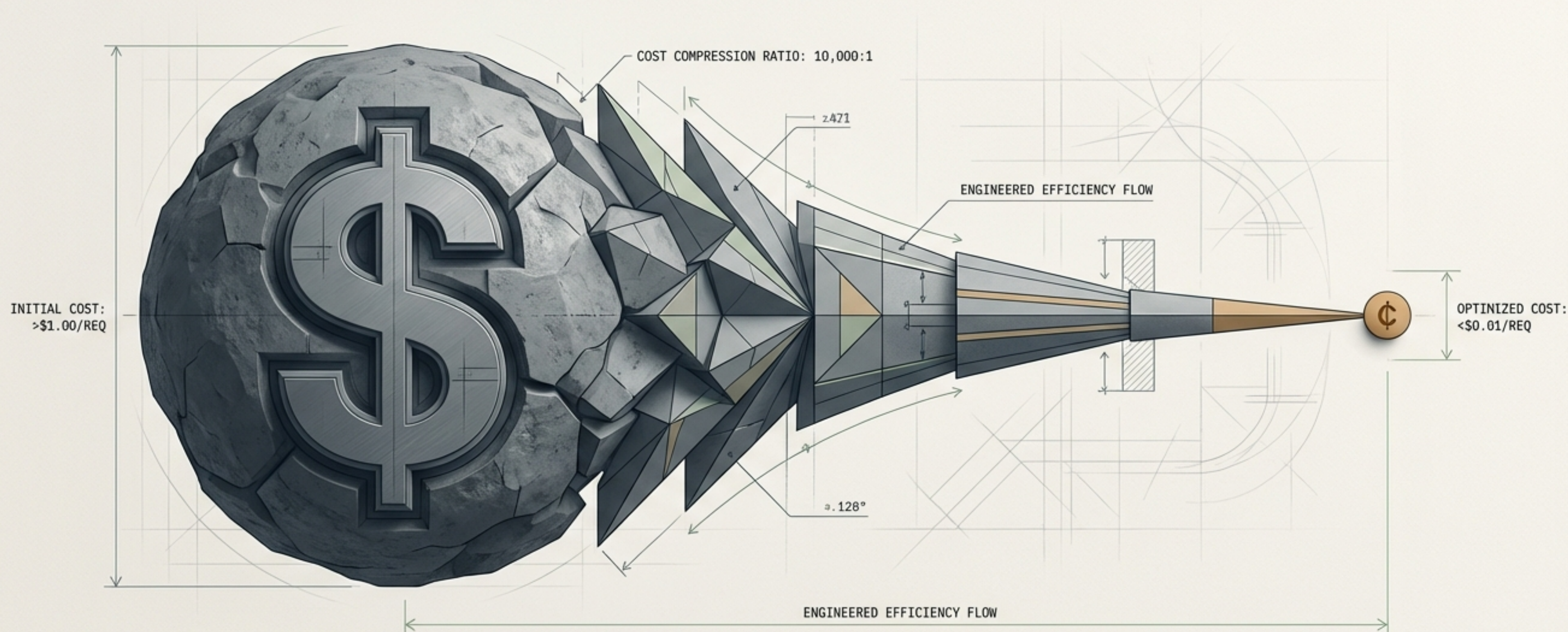
Engineering AI requests from dollars to fractions of a cent

Most teams treat prompts as chat messages.

■ IBM Plex Sans

Treating them as engineering artefacts reduces costs to near zero. ■

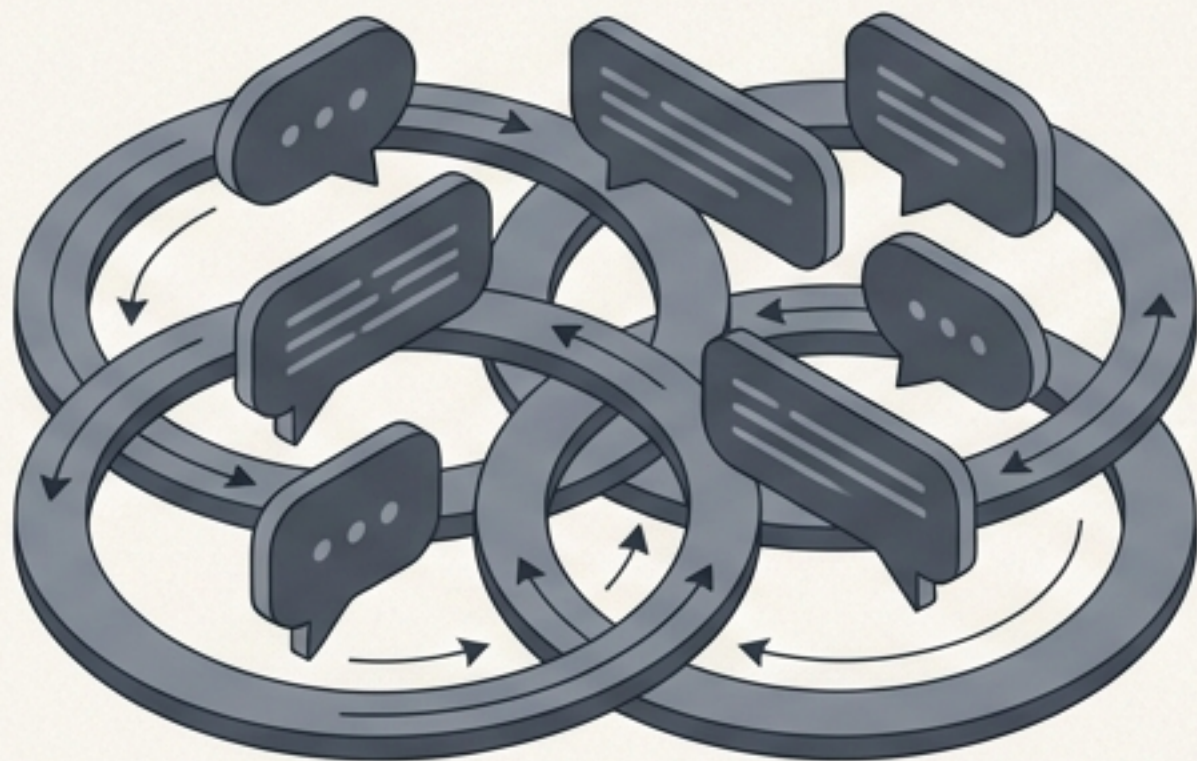
When a single LLM request costs a penny, everything changes. ■



Two fundamentally different approaches to using LLMs

Workflow 1

Let the platform manage it.

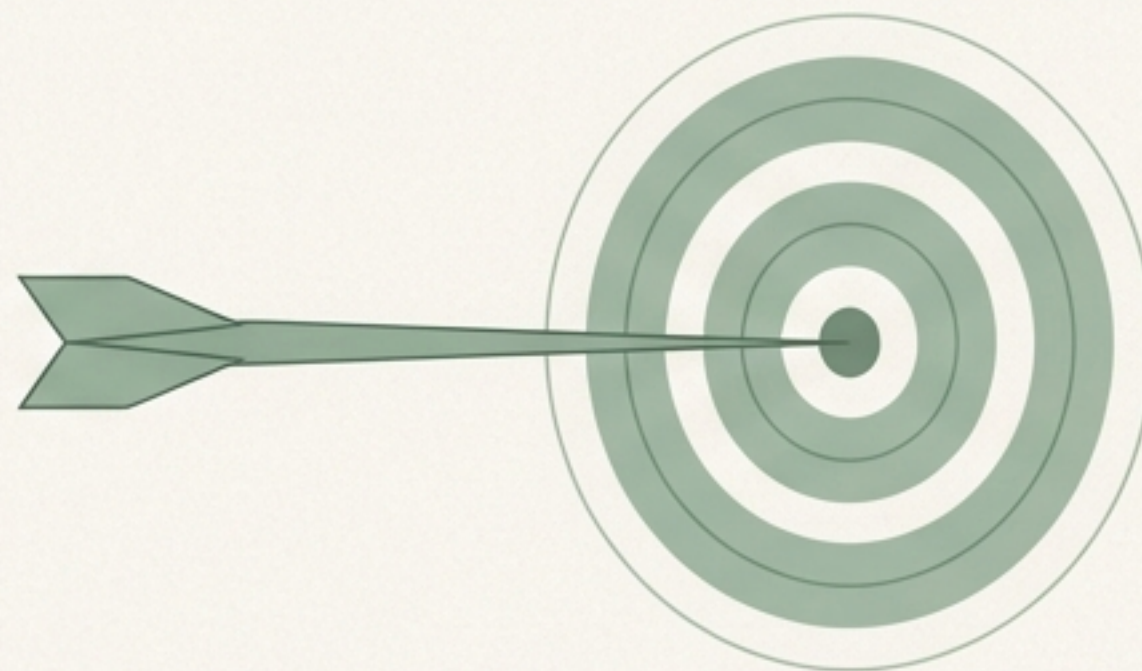


- Multi-turn dialogues
- Platform-optimised context
- Claude Code/Web UI

\$200 / month

Workflow 2

One-shot API requests.



- Developer controls everything
- Self-contained
- No random history

Fractions of a cent per request.

The compounding maths of unoptimised context

A capable model charges \$1–\$3 per 1M input tokens and \$5–\$15 per 1M output tokens.



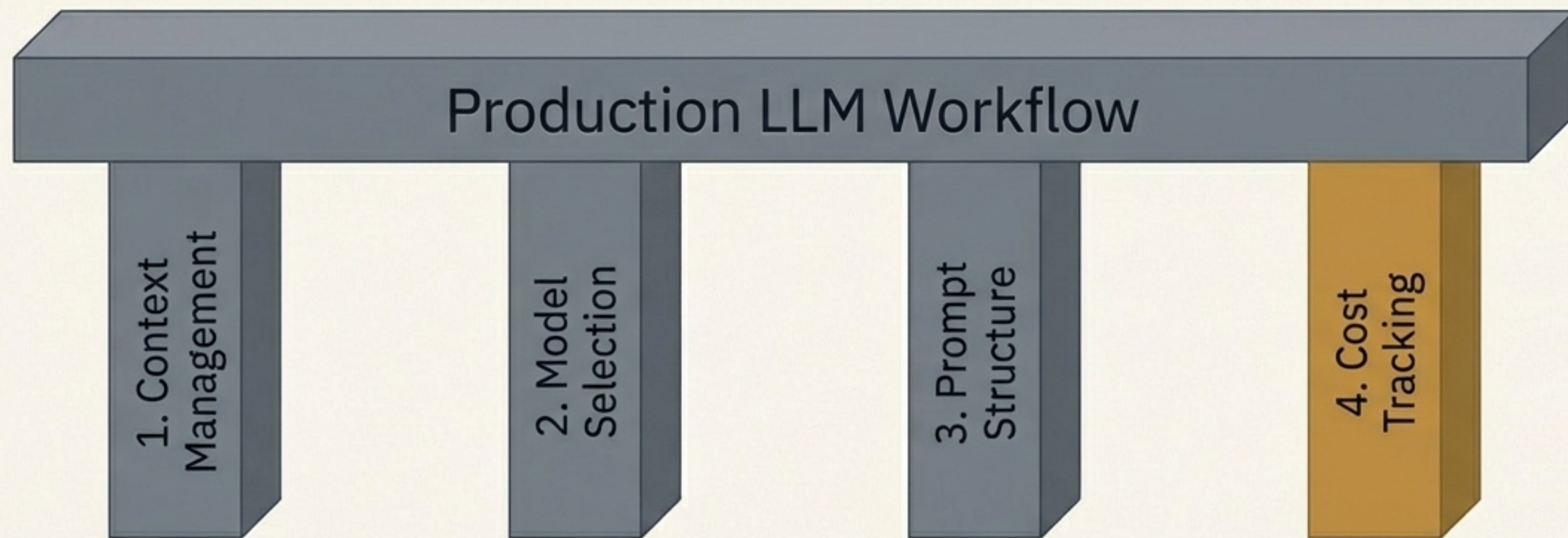
- Unoptimised Request: 50,000 tokens in / 2,000 out
- Engineered Request: 5,000 to 15,000 tokens in / 1,000 out

The model's attention is focused precisely on what matters, not diluted across 40,000 tokens of noise.

High AI costs signal a lack of non-functional engineering

Bloated prompts are the natural state before treating AI requests as system design.

The 4 Pillars of Prompt NFRs:



We must stop sending entire files because we lack the tooling to extract the relevant sections.

Constructing the model's entire reality



Filesystem



Other Projects

Messages Array

System Prompt

Context Files

Question



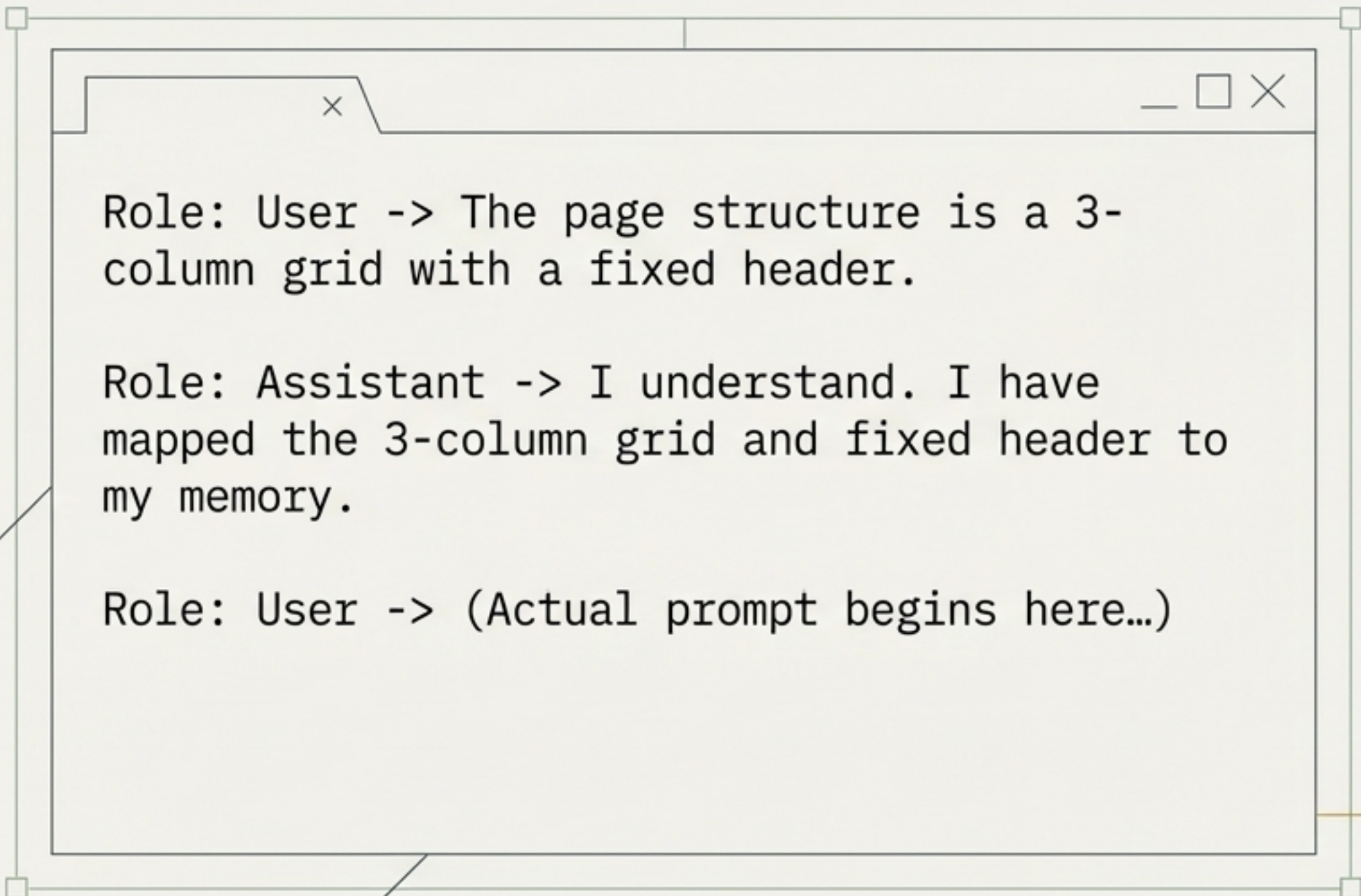
Previous
Conversations

The model sees only what is explicitly placed in the messages array.

Every constraint and context clue is deliberately engineered.

The Golden Rule: Everything the model doesn't need is simply absent. Not hidden, not restricted. Absent.

Designing memory through fabricated conversation history



Role: User -> The page structure is a 3-column grid with a fixed header.

Role: Assistant -> I understand. I have mapped the 3-column grid and fixed header to my memory.

Role: User -> (Actual prompt begins here...)

LLMs are trained on conversations and respond differently to primed dialogue.

Instead of dumping full HTML into a prompt, construct a fake history where the model has already understood the context.

You are designing the model's memory rather than letting it accumulate randomly.

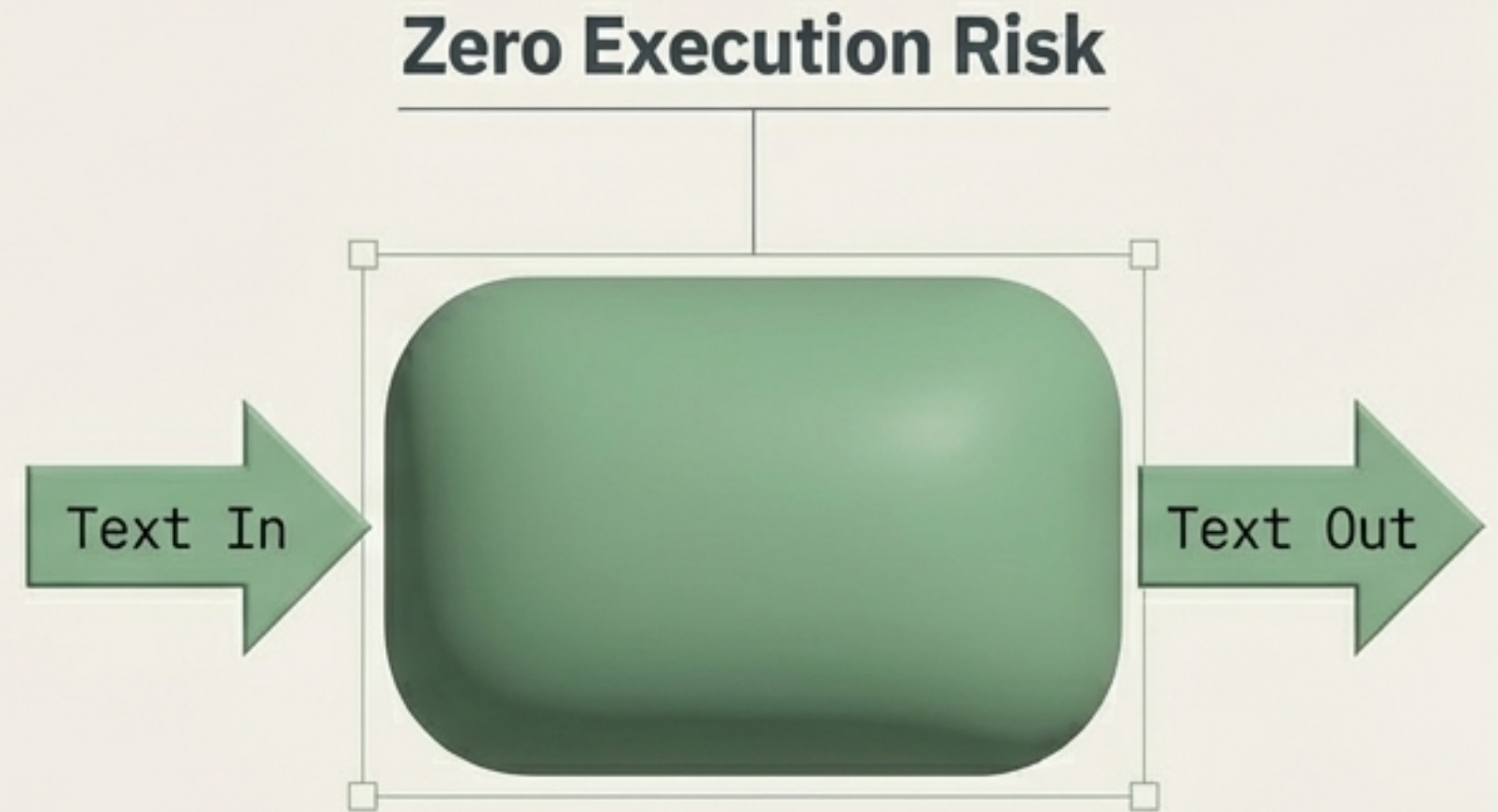
Total cost: Still under 15,000 tokens (under five cents).

The inherent security advantage of the one-shot sandbox



Agentic Tool Access: Every capability is an attack surface vulnerable to prompt injection.

One-Shot API: Inherently sandboxed. Text goes in, text comes out.

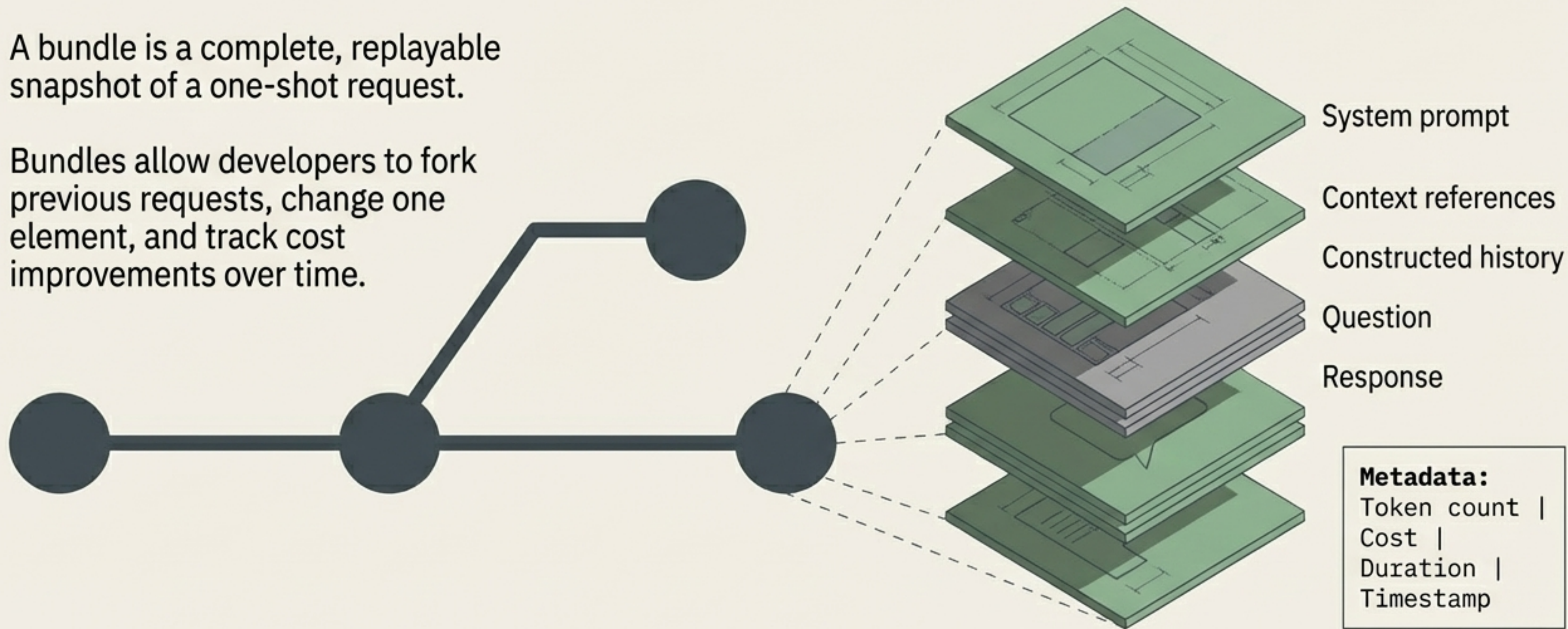


If a response contains malicious instructions, nothing happens because there is no execution environment. The worst case is a bad response you discard.

Execution bundles provide version control for prompts

A bundle is a complete, replayable snapshot of a one-shot request.

Bundles allow developers to fork previous requests, change one element, and track cost improvements over time.

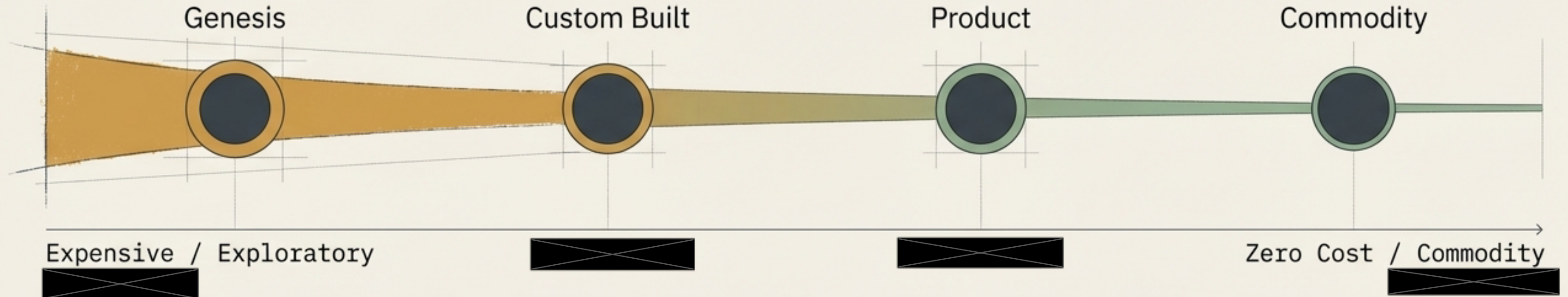


The compression path dictates that every prompt targets zero cost

The Four Stages of Prompt Evolution: Genesis -> Custom Built -> Product -> Commodity.

You cannot optimise what you do not understand. High initial costs are investments in understanding.

The goal is not to start at a penny, but to ensure every request is actively moving toward it.



Stage 1: Genesis is an investment in understanding



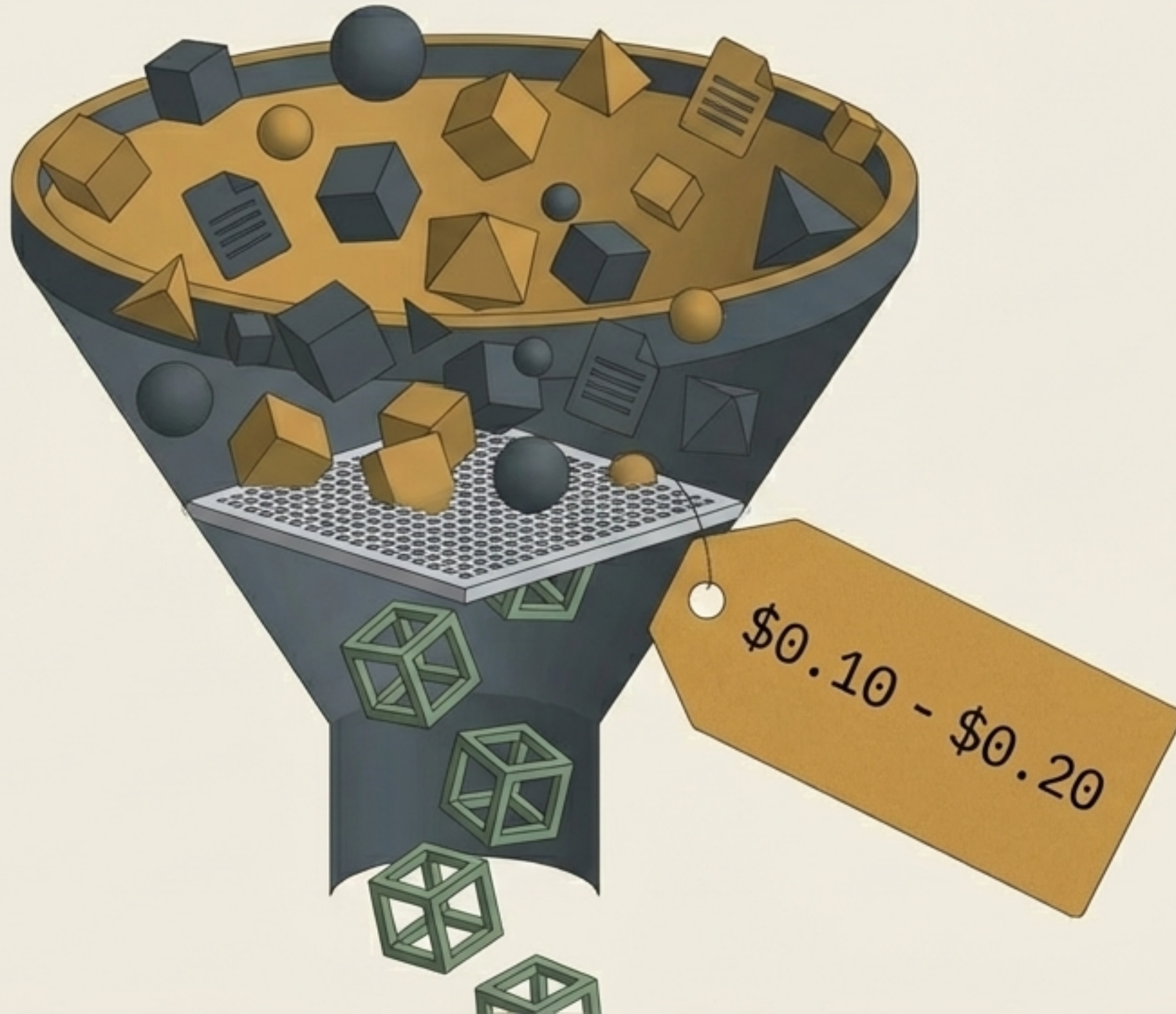
Cost: \$1.00+ per request

Context: 200,000 tokens

Model: The most capable, expensive multimodal models available.

The Goal: Exploration. Discovering the art of the possible and understanding what a good output looks like before attempting to compress the inputs. Save all bundles for analysis.

Stage 2: Custom Built relies on rigorous context curation



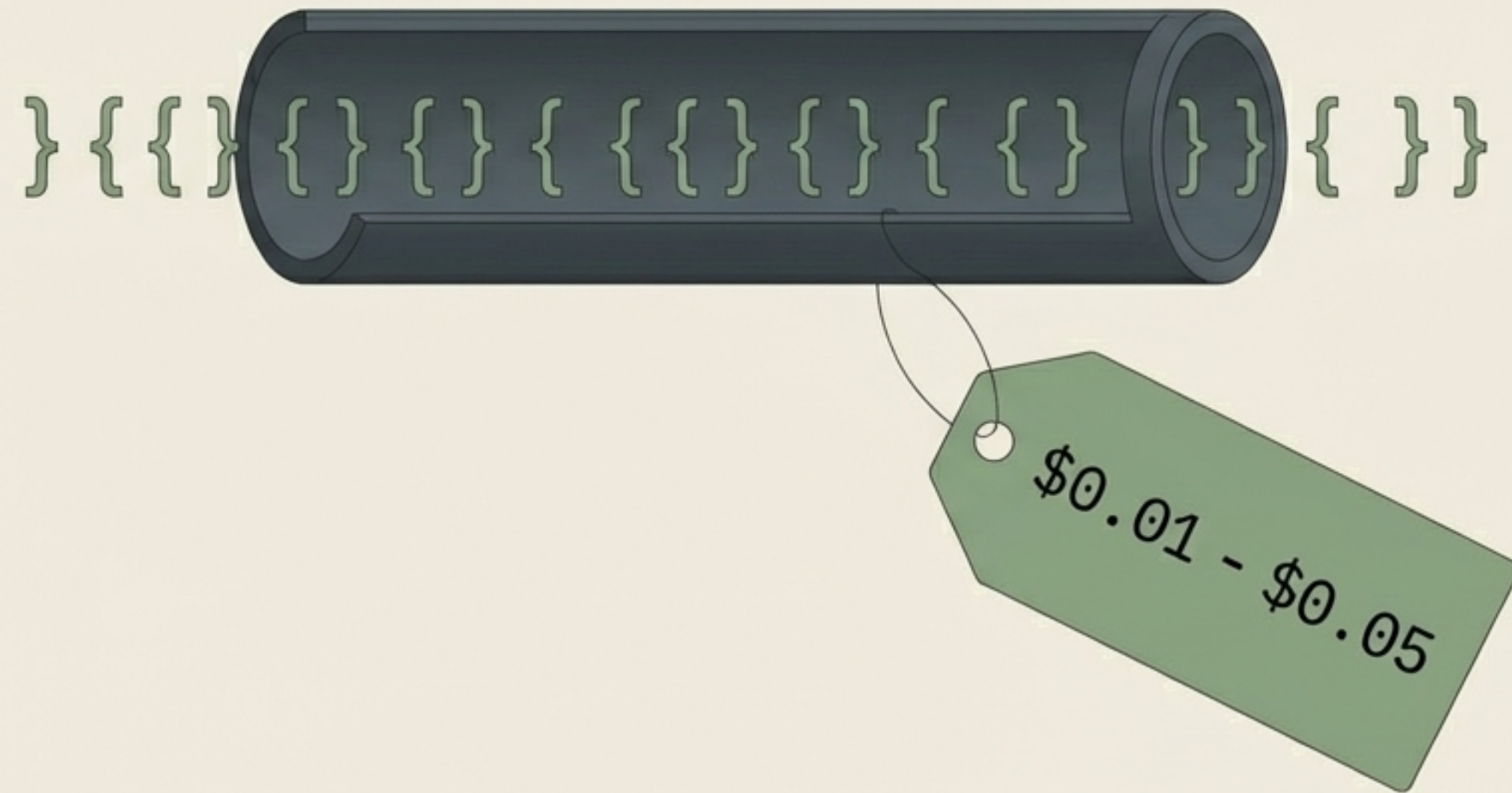
Cost: 10 to 20 cents per request

Context: 10,000 tokens

The Engineering:

- Remove context that did not contribute to the Genesis output.
- Replace full raw documents with structural skeletons.
- Swap expensive multimodal models for text-only models.

Stage 3: Product achieves maximum compression



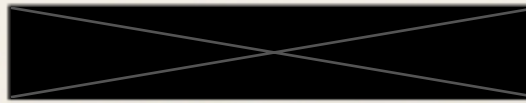
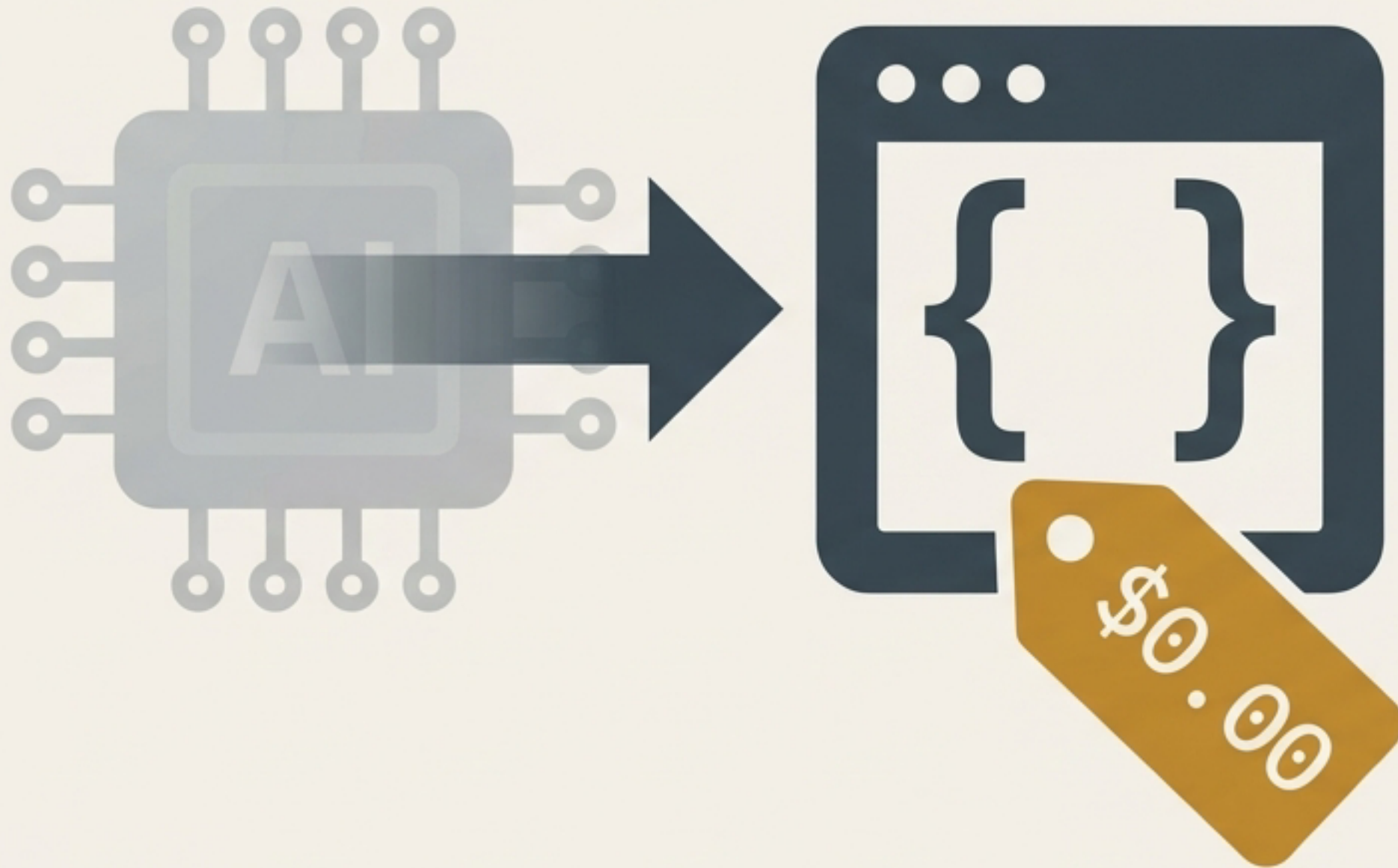
Cost: 1 to 5 cents per request

Context: 3,000 tokens

The Engineering:

- Pre-computed summaries replace raw documents entirely.
- Structured JSON schemas replace verbose natural language instructions.
- Deployment of the cheapest model capable of executing this specific, narrow task.

Stage 4: Commodity removes the LLM entirely



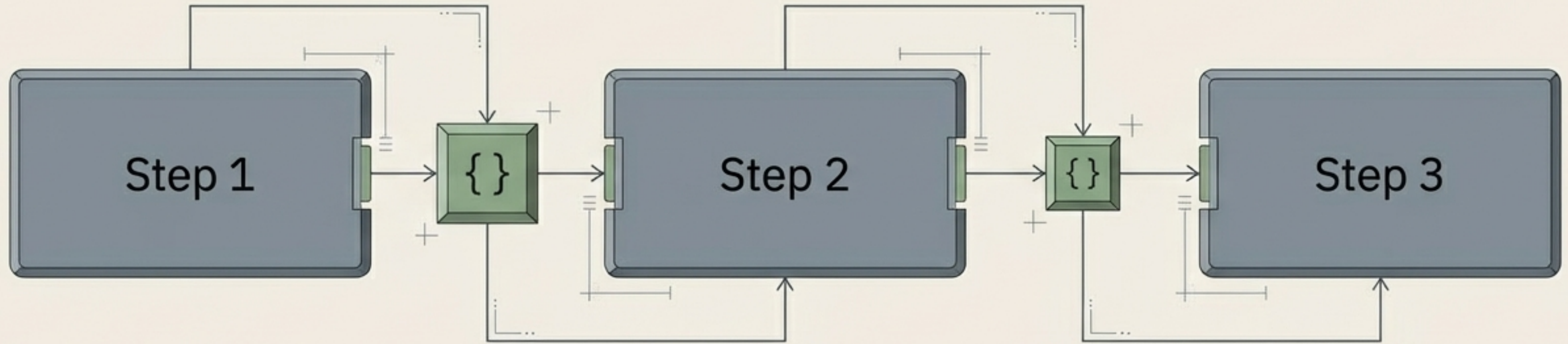
Cost: \$0.00

Context: 0 tokens

The Engineering:

- The rigorous prompt engineering process teaches you enough about the underlying task to write a deterministic script.
- The AI successfully helps write the code that replaces it. Cost becomes zero.

Chaining one-shot requests enables complex orchestration



Currently, one-shot workflows handle individual requests flawlessly. ●

The next evolution is chaining these ultra-cheap requests into **multi-agent** workflows. ●

Because every step is engineered to send only what is needed, complex automated systems will run reliably at a fraction of standard platform costs. Every step remains visible and auditable. ●

The Bellwether Rule of AI Engineering

“ If a known production workflow still costs more than a few cents, you have an engineering opportunity. ”

High costs are not a model problem or a provider problem. They mean the context has not been curated and the NFRs have not been addressed.

The models are now cheap enough that a well-engineered request costs almost nothing. Stop thinking about cost, and start thinking about what to build.